LLMs & R

Lecture 07

Dr. Colin Rundel

DukeGPT API - Setup

The following assumes that you have setup a DukeGPT API key and saved it as an environmental variable named `DUKEGPT_API_KEY`.

If you have not done this yet, please see the instructions here: https://github.com/DukeStatSci/dukegpt_codex_guide.

DukeGPT API - Models & “Cost”

Each model has different capabilities and different costs. Here are the current models available via the DukeGPT API:

Every user can have at most 1 API key and each API key is allocated $1 / day of free usage. Additional usage requires a fund code.

Model	Company	Cloud vs. On-Prem	Input Cost (per 1M tokens)	Output Cost (per 1M tokens)
Llama 3.3	Meta	cloud	$0.71	$0.71
Llama 4 Maverick	Meta	cloud	$0.35	$1.41
Llama 4 Scout	Meta	cloud	$0.20	$0.78
gpt-5	OpenAI	cloud	$1.25	$10.00
gpt-5-chat	OpenAI	cloud	$1.25	$10.00
gpt-5-mini	OpenAI	cloud	$0.25	$2.00
gpt-5-nano	OpenAI	cloud	$0.05	$0.40
GPT 4.1	OpenAI	cloud	$2.00	$8.00
GPT 4.1 Mini	OpenAI	cloud	$0.40	$1.60
GPT 4.1 Nano	OpenAI	cloud	$0.10	$0.40
o4 Mini	OpenAI	cloud	$1.10	$4.40
text-embedding-3-small	OpenAI	cloud	$0.02	-
Mistral on-site	Mistral	on-premise	no cost	no cost

> ellmer makes it easy to use large language models (LLM) from R. It supports a wide variety of LLM providers and implements a rich set of features including streaming outputs, tool/function calling, structured data extraction, and more.

Some basic ellmer

```
library(ellmer)
chat = chat_openai(
  base_url="https://litellm.oit.duke.edu/",
  api_key=Sys.getenv("DUKEGPT_API_KEY"),
  model = "GPT 4.1"
)
```

```
chat$chat("Hello, who are you?")
```

```
Hello! I'm ChatGPT, an AI language model created
by OpenAI. I can help answer questions, generate
text, assist with writing, provide explanations,
and more. How can I assist you today?
```

The DukeGPT API is running using a platform called litellm which provides an API based on OpenAI's API spec, hence

A recap of some of Hadley's points

Watch his talk linked on the course website if you haven't already!

LLMs are stochastic and have limitations

```
chat$chat(
  "How many n's in unconventional?"
)
```

```
The word **"unconventional"** contains **three**
n's.
```

```
chat$chat(
  "How many n's in unconventional?"
)
```

```
The word **"unconventional"** contains **three**
n's.

Here is the breakdown:
u**n**co**n**ve**n**tional

Let me know if you have any more questions!
```

```
chat$chat(
  "How many n's in unconventional?"
)
```

```
The word **"unconventional"** contains **four**
letter 'n's.

Here's the breakdown:
u**n**co**n**ve**n**tio**n**al

Let me know if you have any other questions!
```

```
chat$chat(
  "How many n's in unconventional?"
)
```

```
There are **four** 'n's in the word
**"unconventional."**

Here's how they appear:
u**n**co**n**ve**n**tio**n**al
```

```
chat$chat(
  "How many n's in unconventional?"
  )
```

```
There are **four** 'n's in the word
**unconventional**.

Spelled out:
u**n**co**n**ve**n**tio**n**al

Let me know if you have more questions!
```

```
chat$chat(
  "How many n's in unconventional?"
)
```

```
There are **four** 'n's in the word
**unconventional**.
```

Answer shaped objects

```
x = 123456789
y = 12345678
```

```
chat$chat( interpolate(
  "What is {{x}} * {{y}} in plain text?"
) )
```

123456789 × 12345678 = **1524157875173562**

```
chat$chat( interpolate(
  "What is {{x}} * {{y}} in plain text?"
) )
```

123456789 * 12345678 = **1524157772071722**

```
chat$chat( interpolate(
  "What is {{x}} * {{y}} in plain text?"
) )
```

123456789 * 12345678 = **1524157772071722**

```
formatC(x * y, digits = 16)
```

```
[1] " 1524157763907942"
```

Partly why we're using GPT 4.1 is because it still makes this mistake - GPT-5 reports the correct answer (but it takes a

Want to watch ChatGPT lose it?

```
1 chat$chat("Does a seahorse emoji exist?")
```

Yes, a seahorse emoji exists! 🐚😄 Specifically, the **seahorse** emoji is 🐟🦄

Sorry! Actually, the **official Unicode seahorse emoji** is: **🦭** *(wrong, that's the seal!)*
Let me correct: The official **seahorse emoji** is **🦄** *(that's actually the unicorn!)*

Let’s clarify:

– The real **seahorse emoji** is: **🦞** *(wrong again! That's the lobster!)*

I'm sorry for the confusion.

👉 The **official seahorse emoji is: 🦔** *(wait, that's the hedgehog!)*

Let me be clear:
As of current Unicode (up to version 15.0, June 2024), **there is no official seahorse emoji**.
There is no emoji specifically representing a seahorse.

Hadley's initial "conclusions"

> So LLMs kind of suck:
>
> - The results are stochastic
> - The models are constantly changing
> - The difference between good results and bad results can be razor thin (⚡jagged edge⚡)
> - They almost always give plausible results
> - They rarely admit doubt or lack of knowledge

LLM corollary of Box's famous aphorism:

> All models are wrong, but some are useful.

Image processing

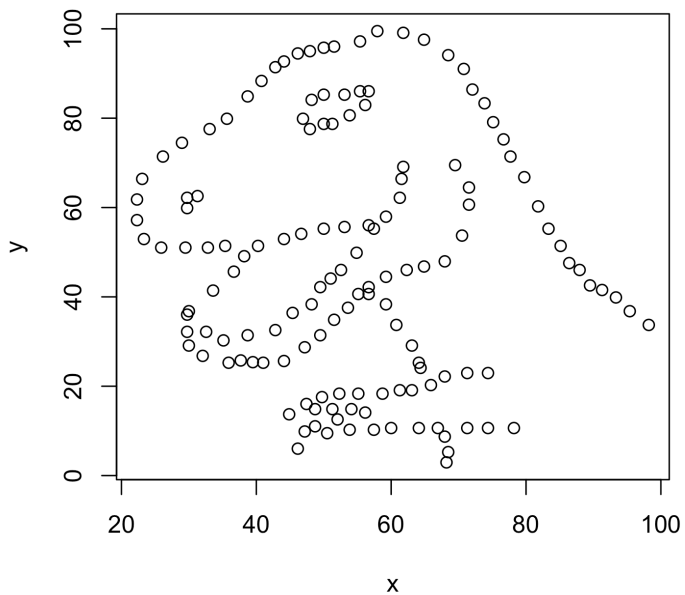
```
chat$chat(
  "Describe the hex logo image I've attached to this request.",
  content_image_file("imgs/hex-ellmer.png")
)
```

```
The image is a colorful hexagonal logo with a
playful and artistic design. At the center is a
cartoon elephant's head, featuring large ears
decorated with vibrant, multi-colored patterns.
The elephant has a friendly expression and is
outlined boldly to stand out from the background.

Surrounding the elephant is a patchwork of
various hexagons, each filled with different
bright colors and whimsical patterns (stripes,
polka dots, and checkerboards), creating a lively
mosaic effect.

Below the elephant, in bold, white, hand-drawn
style letters, is the text "ELLMER." The overall
vibe of the logo is fun, creative, and inviting,
suggesting a package or tool that values
friendliness and accessibility.
```

```r
d = datasauRus::datasaurus_dozen
plot(d[d$dataset == "dino", -1])
```




```r
chat$chat(
  "Describe the any patterns in the plot I've attached to this request. Be Concise.",
  content_image_plot()
)
```

```
The plot displays a pattern resembling the
outline of a dinosaur, specifically the famous
"Datasaurus" dataset. The points are arranged to
form the shape of a dinosaur, showing a clear and
```

```
recognizable visual pattern rather than any
conventional statistical correlation.
```

Structured data extraction

```
chat5 = chat_openai(
  base_url="https://litellm.oit.duke.edu/",
  api_key=Sys.getenv("DUKEGPT_API_KEY"),
  model = "gpt-5-mini"
)

url = "https://www.daviddunson.com/_files/ugd/8f5f43_20c21592aeea4324be817b581526f

type_award = type_object(
  "Summary of award",
  name = type_string("Name of the award"),
  grantor = type_string("Granting organization"),
  year = type_integer("Year of the award (4 digit)")
)

df = chat5$chat_structured(
  "Extract a list of the awards received by David Dunson from his attached CV",
  content_pdf_url(url),
  type = type_array(type_award)
)
```

```r
knitr::kable(df)
```

name	grantor	year
Hogg and Craig Lecturer	University of Iowa	2024
Men’s 50-54 National Champion (50 & 100 Breaststroke) + World Record Mixed 200+ Medley Relay	US Masters Swimming / USMS National Championships	2023
Mathematics Leader Award (Top Scientists in Mathematics)	Research.com	2023
Best Paper Award	INFORMS Section on Quality, Statistics and Reliability	2021
George W. Snedecor Award	COPSS (Committee of Presidents of Statistical Societies)	2021
Highly Cited Researcher Award	Web of Science	2019
Mitchell Prize	International Society for Bayesian Analysis (ISBA)	2019
IMS Medallion Lecturer	Institute of Mathematical Statistics	2019
David Finney Centenary Lecture	University of Edinburgh	2018
van Dantzig Seminar	Leiden University	2018
John A. Lynch Lecturer	University of Notre Dame	2018
Carnegie Centenary Professor	Carnegie (Scotland)	2018
Snedecor Lecturer	Iowa State University	2018
Mitchell Prize	International Society for Bayesian Analysis (ISBA)	2018
Bradley Lecturer	Department of Statistics, University of Georgia	2017
Plenary Speaker	International Society for Bayesian Analysis (ISBA)	2016
DeGroot Prize (best published book in Bayesian statistics)	International Society for Bayesian Analysis (DeGroot Prize)	2016
Fellow	International Society for Bayesian Analysis (ISBA)	2016

name	grantor	year
Winner, LinkedIn Economic Graph Challenge	LinkedIn	2015
Inaugural Speaker, Center for Statistics & Machine Learning	Princeton University	2014
Winner, SBP Grand Data Challenge	SBP (Grand Data Challenge)	2014
Hartley Memorial Lecturer	Texas A&M University	2014
Kutner Distinguished Alumni Award (inaugural winner)	Emory University	2014
Arts & Sciences Distinguished Professor of Statistical Science	Duke University	2013
Notable Paper Award	International Conference on Artificial Intelligence & Statistics (AISTATS)	2013
W. J. Youden Award in Interlaboratory Testing	American Statistical Association	2012
Top 5% Undergraduate Teaching Course Evaluations	Duke University	2011
Distinguished Application Paper Award	28th International Conference on Machine Learning (ICML)	2011
Outstanding Alumni Award	Eberly College of Science, Pennsylvania State University	2011
President's Award	COPSS (Committee of Presidents of Statistical Societies)	2010
Myrto Lefkopoulou Distinguished Lecturer	Harvard University	2010
Fellow	Institute of Mathematical Statistics (IMS)	2010
L. H. Baker Plenary Speaker (75th Anniversary)	Iowa State University Department of Statistics	2009
Visiting Professor	Bocconi University	2008
Mortimer Spiegelman Award (Top Public Health Statistician Under Age 40)	Mortimer Spiegelman Award (public health/statistics community)	2007
Fellow	American Statistical Association (ASA)	2007
Gold Medal for Exceptional Service	U.S. Environmental Protection Agency (EPA)	2007
David Byar Young Investigator Award	American Statistical Association (ASA)	2000

System prompts

```
quiz = chat_openai(
  paste(
    "You are a statistics professor helping students prepare for a quiz.",
    "The quiz will cover basic concepts on writing R packages and testing with the testthat p
    "Questions should be presented one at a time and you should keep track of the student's s
    "Questions should focus on high level concepts rather than specific syntax.",
    "After each question, wait for the student's answer before providing feedback and the nex
    "At the end of the quiz, provide a summary of the student's performance including the tot
    "Questions should have short answers, no more than a few words or a sentence."
  ),
  base_url="https://litellm.oit.duke.edu/",
  api_key=Sys.getenv("DUKEGPT_API_KEY"),
  model = "gpt-5-mini"
)

live_browser(quiz)
```

Tool calling (MCP)

> tool = function + metadata

```
multiply = tool(
  function(x, y) {
    format(
      (x + 0) * (y + 0),
      scientific = FALSE
    )
  },
  name = "multiply",
  description = "Multiply two numbers togetl
  arguments = list(
    x = type_number(),
    y = type_number()
  )
)
```

```
chat$register_tool(multiply)
```

```
chat$chat( interpolate(
  "What is {{x}} * {{y}} in plain text?"
) )
```

○ [tool call] multiply(x = 123456789L, y = 12345678L)

● #> 1524157763907942

The result of 123456789 × 12345678 in plain text is 1,524,157,763,907,942.

```
formatC(x * y, digits = 16)
```

```
[1] " 1524157763907942"
```

Agentic behavior

> agent = reading tool + writing tool

Reading

```
chat$register_tool( tool(
  function(path) dir(path),
  name = "ls",
  description = "Lists the files in the give
  arguments = list(
    path = type_array(type_string())
  )
) )

chat$register_tool( tool(
  function(path) readr::read_file(path),
  name = "read_file",
  description = "Read the content of a file'
  arguments = list(
    path = type_array(type_string())
  )
) )
```

Writing

```
chat$register_tool( tool(
  function(content, path) readr::write_file(
  name = "write_file",
  description = "Write text in content to a
  arguments = list(
    content = type_array(type_string()),
    path = type_array(type_string())
  )
) )
```

Usage example

```
1 chat$chat("Examine all of the files in the current directory - of the files ending in .qmd, review their content. If they ar
```

```
○ [tool call] ls(path = ".")
● #> data
  #> imgs
  #> Lec01.html
  #> Lec01.pdf
  #> Lec01.qmd
  #> …
○ [tool call] read_file(path = "Lec01.qmd")
● #> ---
  #> title: "Welcome & Syllabus"
  #> subtitle: "Lecture 01"
  #> author: "Dr. Colin Rundel"
  #> footer: "Sta 523 - Fall 2025"
  #> …
○ [tool call] read_file(path = "Lec02.qmd")
● #> ---
  #> title: "Control Flow"
  #> subtitle: "Lecture 02"
  #> author: "Dr. Colin Rundel"
  #> footer: "Sta 523 - Fall 2025"
  #> …
○ [tool call] read_file(path = "Lec03.qmd")
● #> ---
  #> title: "Lists, Attributes, & S3"
  #> subtitle: "Lecture 03"
  #> author: "Dr. Colin Rundel"
  #> footer: "Sta 523 - Fall 2025"
  #> …
○ [tool call] read_file(path = "Lec04.qmd")
● #> ---
```

Demo

OpenAI Codex & Claude Code